

Strong disorder and noise sensitivity for the 2D Stochastic Heat Equation

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Outline

1. Stochastic Heat Equation
2. Critical regime
3. Directed polymers
4. Noise sensitivity
5. Strong disorder
6. Conclusions

In a nutshell

Stochastic Heat Equation

ill-defined in space dim. $d \geq 2$

$$\partial_t u(t, x) = \underbrace{\Delta_x u(t, x)}_{\sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} u(t, x)} + \underbrace{\beta \xi(t, x)}_{\text{"space-time white noise"} \atop \text{Gaussian cov. } \delta(t - t') \delta(x - x')} \cdot u(t, x)$$

There is a **non-trivial solution** in critical space dimension $d = 2$

Critical 2D Stochastic Heat Flow (SHF)

We study its properties: **noise sensitivity** and **strong disorder**

Singular SPDEs



Statistical Mechanics

The role of dimension

$$\partial_t u(t, x) = \Delta_x u(t, x) + \beta \xi(t, x) u(t, x)$$

Space-time blow-up $\tilde{u}(t, x) := u(\delta^2 t, \delta x)$

$$\partial_t \tilde{u}(t, x) = \Delta_x \tilde{u}(t, x) + \delta^{\frac{2-d}{2}} \beta \tilde{\xi}(t, x) \tilde{u}(t, x)$$

As $\delta \downarrow 0$ the noise strength $\begin{cases} \text{vanishes} & d < 2 & \text{sub-critical} \\ \text{unchanged} & d = 2 & \textbf{critical} \\ \text{diverges} & d > 2 & \text{super-critical} \end{cases}$

We will henceforth focus on **critical dimension** $d = 2$

Regularization

Regularized noise $\xi_1(t, x)$ $\rightsquigarrow$ well-defined solution $u_1(t, x)$

(say: discretized on scale 1)

$$\partial_t u_1(t, x) = \Delta_x u_1(t, x) + \beta \xi_1(t, x) u_1(t, x) \quad u_1(0, x) \equiv 1 \text{ (say)}$$

Diffusive rescaling $u_N(t, x) := u_1(Nt, \sqrt{N}x)$ (large-scale of u_1)

$$\partial_t u_N(t, x) = \Delta_x u_N(t, x) + \beta \xi_N(t, x) u_N(t, x) \quad u_N(0, x) \equiv 1$$

$$\xi_N(t, x) := N \xi_1(Nt, \sqrt{N}x) \xrightarrow[N \rightarrow \infty]{d} \text{white noise } \xi(t, x)$$

Convergence?

Two interpretations for $u_N(t, x)$

- ▶ Behavior of **regularized SHE solution** $u_1(t, x)$ on large diffusive scales
- ▶ Approximation of **ill-defined SHE** by small-scale regularization of white noise

Convergence of $u_N(t, \varphi) = \int_{\mathbb{R}^2} u_N(t, x) \varphi(x) dx$? $[\varphi \in C_c: \varphi \geq 0, \int \varphi = 1]$

NO! Theorem

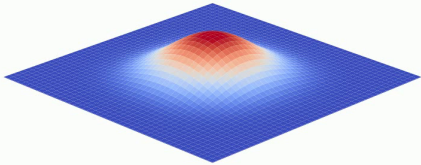
[Berger C. Turchi 25]

For every fixed $\beta > 0$ and $t > 0$: $u_N^\beta(t, \varphi) \xrightarrow[N \rightarrow \infty]{d} 0$ (finer results later)

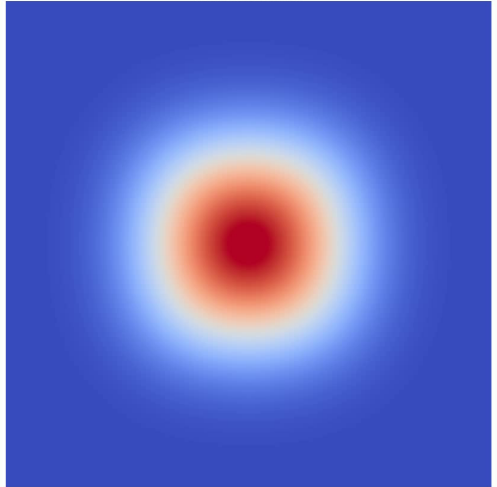
Discretization: $\beta = 0$ (deterministic)

(by Q. Berger)

Density after n=14448 steps



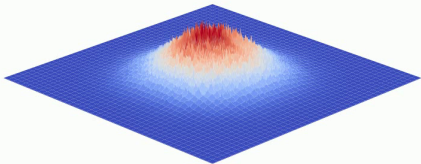
Heat map after n=14448 step



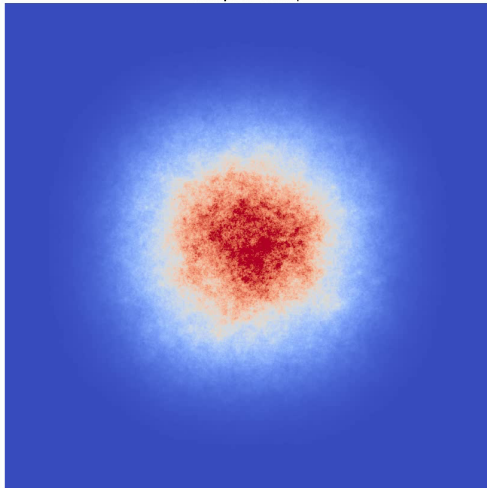
Discretization: β very small

(by Q. Berger)

Density after $n=14448$ steps

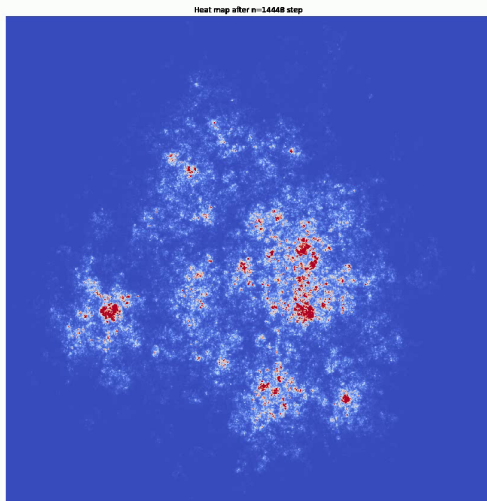
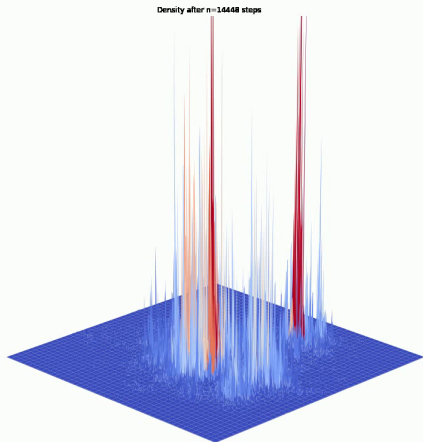


Heat map after $n=14448$ step



Discretization: β “critical” (still small!)

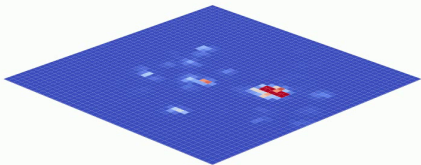
(by Q. Berger)



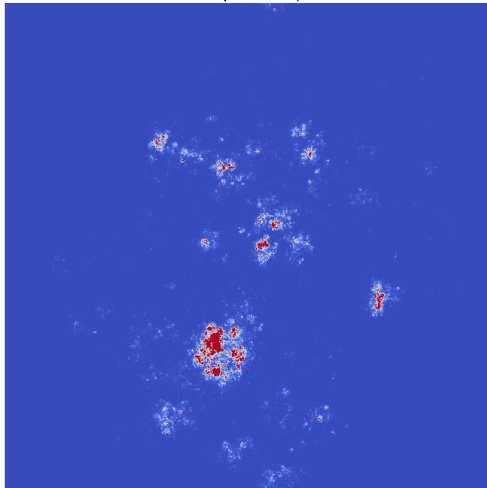
Discretization: β “supercritical”

(by Q. Berger)

Density after $n=14448$ steps



Heat map after $n=14448$ step



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Renormalization

Why $u_N^\beta(t, \varphi) \rightarrow 0$ for fixed $\beta > 0$?

- ▶ Mean is constant $\mathbb{E}[u_N^\beta(t, \varphi)] = \int \varphi \equiv 1$ [martingale]
- ▶ 2nd moment explodes $\mathbb{E}[u_N^\beta(t, \varphi)^2] \rightarrow \infty \quad \forall \text{ fixed } \beta > 0$

Tame moment explosion: critical tuning $\beta^2 = \beta_N^2 \sim \frac{\pi}{\log N} \rightarrow 0$

- ▶ $\mathbb{E}[u_N^{\beta_N}(t, \varphi)^2] \rightarrow K_t(\varphi, \varphi) > 0$ [Bertini Cancrini 98] [C.S.Z. 19]
- ▶ q -th moment converges for all $q > 2$ [C.S.Z. 19] [Gu Quastel Tsai 21]
Fast growth $\gtrsim \exp \exp(q)$

The critical 2D Stochastic Heat Flow

Theorem

[C.S.Z. 23]

Critical window $\beta_N^2 = \frac{\pi}{\log N} \left(1 + \frac{\vartheta}{\log N} \right)$ for some $\vartheta \in \mathbb{R}$

Convergence of u_N to a **unique** and **non-trivial limit** SHF^ϑ

$$\left(u_N^{\beta_N}(t, x) dx \right)_{t \geq 0} \xrightarrow[N \rightarrow \infty]{d} \left(\text{SHF}^\vartheta(t, dx) \right)_{t \geq 0}$$

$\text{SHF}^\vartheta =$ critical 2D **Stochastic Heat Flow (SHF)** = process of **random measures** on $\mathbb{R}^2$

Key properties of SHF

► $\text{SHF}^{\vartheta}(t, dx) \neq \text{Gaussian}$ (positive random measure)

► $\text{SHF}^{\vartheta}(t, dx) \neq \text{GMC} \simeq \exp(\text{Gaussian})$ on $\mathbb{R}^2$ [CSZ 23]

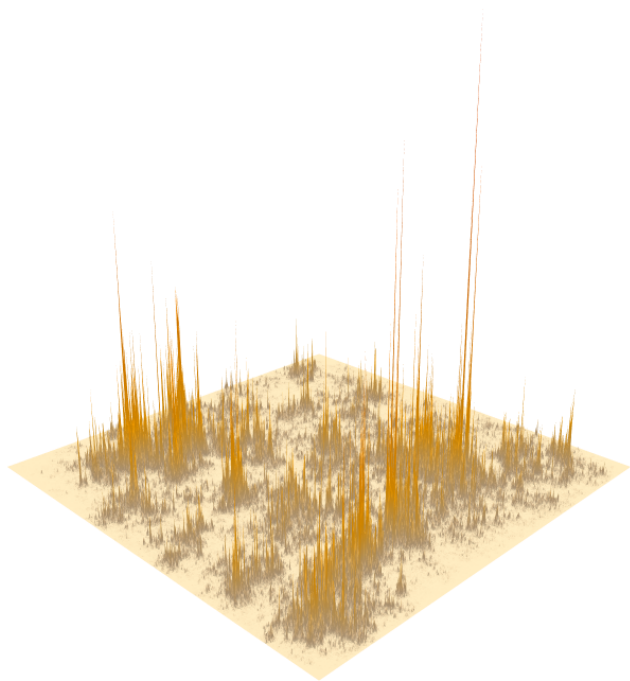
But **conditional GMC structure** on path space [Clark–Tsai 25]

► Moment formulas [CSZ 19] [Gu–Quastel–Tsai 21] [Ganguly–Nam 25]

$\rightsquigarrow$ [Cosco, Nakajima, Zeitouni] [Lygkonis, Zygouras] [Liu, Zygouras] ...

► Axiomatic characterisation by independence and moments [Tsai 24]

► Diffusive rescaling $\text{SHF}^{\vartheta}(\alpha t, d(\sqrt{\alpha}x)) \stackrel{d}{=} \alpha \text{SHF}^{\vartheta+\log \alpha}(t, dx)$

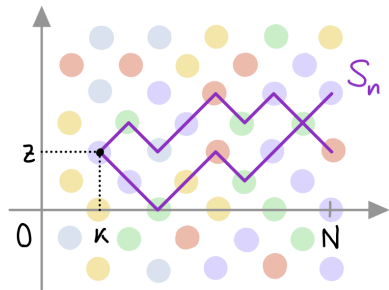


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Directed polymers partition functions

- ▶ $S = (S_n)_{n \geq 0}$ simple random walk on $\mathbb{Z}^d$
- ▶ Independent Gaussians $\omega(n, x) \sim \mathcal{N}(0, 1)$
- ▶ $H(S, \omega) := \sum_{n=k+1}^N \omega(n, S_n)$



Partition Functions

$$(k \in \mathbb{N}, z \in \mathbb{Z}^d)$$

$$Z_{N,\beta}^{\omega}(k, z) = \mathbb{E} \left[e^{\beta H(S, \omega) - \frac{1}{2} \beta^2 (N-k)} \mid S_k = z \right]$$

$$\mathbb{E} [Z_{N,\beta}^{\omega}] = 1$$

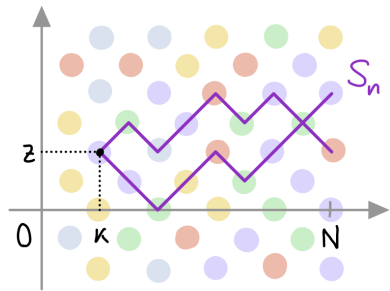
Partition functions and SHE

Partition functions $Z_{N,\beta}^{\omega}(k,z)$ are solutions of
discretised Stochastic Heat Equation

Diffusive rescaling (+ time reversal)

$$u_N(t,x) := Z_{N,\beta}^{\omega}(N(1-t), \sqrt{N}x)$$

$$\begin{cases} \partial_t u_N(t,x) = \Delta_x u_N(t,x) + \underbrace{\beta \xi_N(t,x)}_{\text{discretized white noise}} u_N(t,x) \\ u_N(0,x) \equiv 1 \end{cases} \quad (\text{disc-SHE})$$



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SHF and white noise

Regularized Stochastic Heat Equation

$$\partial_t u_N(t, x) = \Delta_x u_N(t, x) + \beta_N \xi_N(t, x) u_N(t, x) \quad u_N(0, \cdot) \equiv 1$$

Removing the regularization ($N \rightarrow \infty$)

$$\xi_N \rightarrow \xi \text{ white noise} \quad u_N \rightarrow \text{SHF}^\vartheta \quad \text{critical } \beta = \beta_N \rightarrow 0$$

Naively
$$\partial_t \text{SHF}^\vartheta(t, dx) = \Delta_x \text{SHF}^\vartheta(t, dx) + 0 \quad ???$$

Does SHF^ϑ solve a stochastic PDE? Joint convergence of (u_N, ξ_N) ?

No SPDE for SHF

Theorem

[C.–Donadini 25]

$$(u_N, \xi_N) \xrightarrow[N \rightarrow \infty]{d} (\text{SHF}^\vartheta, \xi) \quad \text{SHF}^\vartheta \text{ and } \xi \text{ independent}$$

Even though u_N is a **function** of ξ_N , dependence is completely lost as $N \rightarrow \infty$

SHF^ϑ cannot solve a SPDE driven by ξ

Theorem

[Gu-Tsai 25]

SHF^ϑ is a “black noise” (Tsirelson)

Noise sensitivity

We show that u_N is **sensitive** to small perturbations of the noise ξ_N

$f_N(\omega) := Z_{N,\beta}^\omega$ is **sensitive** to small perturbations of the environment ω

Resample each $\omega(n, z)$ with probability $\varepsilon \rightsquigarrow \omega^\varepsilon = (\omega^\varepsilon(n, z))_{n,z}$

Enhanced **Noise Sensitivity** [Benjamini–Kalai–Schramm 99] [C.–Donadini 25]

$$\forall \varepsilon \in (0, 1) \quad \text{Corr}[\phi(f_N(\omega)), \psi(f_N(\omega^\varepsilon))] \xrightarrow{N \rightarrow \infty} 0 \quad \forall \phi, \psi \in C_b^\infty$$

$$\implies \text{Corr}[\phi(f_N(\omega)), P(\omega)] \xrightarrow{N \rightarrow \infty} 0 \quad \forall P \text{ polynomial}$$

$$\rightsquigarrow f_N(\omega) \text{ asymptotic independent of } \xi_N$$

Influences

Chaos exp. $f_N(\omega) = \mathbb{E}[f_N] + \sum_{d=1}^{\infty} f_N^{(d)}(\omega)$ $f_N^{(d)}(\omega) = \text{degree } d \text{ polynomial}$

Enhanced Noise Sensitivity: $\|\phi(f_N^{(d)}(\omega))\|_{L^2} \xrightarrow{N \rightarrow \infty} 0 \quad \forall \text{ fixed } d$

Gradient

Influence

[C.–Donadini 25]

$$\delta_i f(\omega) := f(\omega) - \mathbb{E}_i[f(\omega)]$$

$$\text{Inf}_i[f(\omega)] := \left\| \delta_i f(\omega) \right\|_{L^1_{\neq i}} \left\| \right\|_{L^2_i}$$

BKS criterion

[Benjamini–Kalai–Schramm 99] [C.–Donadini 25]

Enhanced Noise Sensitivity if $\sum_i \text{Inf}_i[f_N(\omega)]^2 \xrightarrow{N \rightarrow \infty} 0$

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Supercritical regime

Regularized Stochastic Heat Equation

$$\partial_t u_N(t, x) = \Delta u_N(t, x) + \beta_N \xi_N(t, x) u_N(t, x) \quad u_N(0, \cdot) \equiv 1$$

$$\xi_N \rightarrow \xi \text{ white noise} \quad u_N \rightarrow \text{SHF}^{\vartheta} \quad \text{critical } \beta = \beta_N \rightarrow 0$$

$$\beta^2 = \frac{\pi}{\log N} \left(1 - \frac{\vartheta}{\log N} \right)^{-1} = \frac{\pi}{\log N - \vartheta}$$

Strong disorder regime

(supercritical)

Any regime of $\vartheta \rightarrow \infty$

$$1 \ll \vartheta \leq \log N - O(1)$$

Main result: local extinction

Theorem

[Berger C. Turchi 25]

$$u_N(t, \varphi) \xrightarrow[N \rightarrow \infty]{d} 0 \quad \text{as soon as } \vartheta \rightarrow \infty \quad \text{SHF}^\vartheta(t, \varphi) \xrightarrow[\vartheta \rightarrow \infty]{d} 0$$

(incl. fixed $\beta > 0$)

We establish quantitative bounds via

Fractional moments

Truncated mean

$$\mathbb{E}[u_N(t, \varphi)^\gamma] \quad \text{with } \gamma \in (0, 1)$$

$$\mathbb{E}[u_N(t, \varphi) \wedge 1] = \mathbb{P}(u_N(t, \varphi) > U(0, 1))$$

We focus for simplicity on $t = 1$ and

$$\varphi(\cdot) = B_R(\cdot) = \frac{1}{\pi R^2} \mathbb{1}_{|\cdot| \leq R}$$

Quantitative bounds

Theorem

[Berger C. Turchi 25]

Fix any $R > 0$. There are c, C such that

$$c \exp(-C e^{\vartheta}) \leq \mathbb{E}[u_N(1, B_R) \wedge 1] \leq C \exp(-c e^{\vartheta})$$

Note that

$$e^{\vartheta} = e^{-\pi/\beta^2} N = O(1) \text{ for } \beta = \beta_N \text{ critical}$$

$\forall \varepsilon \in (0, 1)$ there are $c_\varepsilon, C_\varepsilon$ such that

$$c_\varepsilon \exp(-C e^{\vartheta}) \leq \mathbb{P}(u_N(1, B_R) \geq \varepsilon) \leq C_\varepsilon \exp(-c e^{\vartheta})$$

LB: 2nd moment method

UB: coarse-graining + change of measure

Spatial scale of mass escape

For fixed $c > 0$ define

$$R_{\vartheta}^c := \exp(c e^{\vartheta}) = \exp(c e^{-\pi/\beta^2} N) \quad = O(1) \text{ for } \beta = \beta_N \text{ critical}$$

Theorem

[Berger C. Turchi 25]

There are $0 < c_- \leq c_+ < \infty$ such that

$$u_N(1, R_{\vartheta}^c) \xrightarrow[N \rightarrow \infty]{d} \begin{cases} 0 & \text{if } c < c_- \\ 1 & \text{if } c > c_+ \end{cases}$$

Conjecture: sharp transition $c_- = c_+$

Main results for directed polymers

Theorem

[Berger C. Turchi 25]

$$c \exp(-C e^{\vartheta}) \leq \mathbb{E}[Z_{N,\beta}^{\omega}(B_{R\sqrt{N}}) \wedge 1] \leq C \exp(-c e^{\vartheta}) = C \exp(-c e^{-\pi/\beta^2} N)$$

Free energy (point-to-plane)

[Berger Lacoin 17]

$$F(\beta) := \lim_{N \rightarrow \infty} \frac{1}{N} \log Z_{N,\beta}(0) = \exp\left(-\frac{\pi + o(1)}{\beta^2}\right) \quad \text{as } \beta \downarrow 0$$

Theorem

[Berger C. Turchi 25] [Berger Nakajima 26]

$$-C \exp\left(-\frac{\pi}{\sigma^2(\beta)}\right) \leq F(\beta) \leq -c \exp\left(-\frac{\pi}{\sigma^2(\beta)}\right) \quad \text{with } \sigma^2(\beta) := e^{\beta^2} - 1$$

Proof of the UB: coarse-graining

$$\mathbb{E}[u_N^\beta(1, B_1) \wedge 1] = \mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq C \exp(-c e^\vartheta)$$

► Coarse-graining argument: $\begin{cases} \text{reduce } \vartheta \rightarrow \infty \text{ to fixed (large) } \vartheta > 0 \\ \text{replace } \exp(-c e^\vartheta) \text{ by any } f(\vartheta) \rightarrow 0 \end{cases}$

Key bound

$$\mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq \frac{C}{\vartheta} \quad \text{for fixed } \vartheta \in \mathbb{R}$$

Proof of the UB: change of measure

Change of scale: $\mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq \frac{C}{\epsilon} \mathbb{E}[Z_{N,\beta}(B_{\sqrt{\epsilon N}}) \wedge 1] \quad \forall \epsilon \in (0,1)$

Size-biased law $\tilde{\mathbb{P}}(d\omega) := Z(\omega) \mathbb{P}(d\omega) \quad Z(\omega) := Z_{N,\beta}(B_{\sqrt{\epsilon N}})$

Change of measure

cf. [Junk-Lacoin]

$$\mathbb{E}[Z \wedge 1] \leq \mathbb{P}(A) + \tilde{\mathbb{P}}(A^c) \quad \text{for any event } A$$

Goal: find some event A which is atypical for $\mathbb{P}$ but typical for $\tilde{\mathbb{P}}$

Optimal choice $A = \{Z > 1\}$ (but we don't know Z)

Proof of the UB: choice of a proxy

- ▶ Choose a “nice” r.v. X with $\mathbb{E}[X] = 0$ and $\tilde{\mathbb{E}}[X] \gg 1$ ($X \approx Z - 1$)
- ▶ Take $A = \{X > \frac{1}{2} \tilde{\mathbb{E}}[X]\}$ $\rightsquigarrow$ $\mathbb{P}(A) \leq 4 \frac{\text{Var}[X]}{\tilde{\mathbb{E}}[X]^2}$ $\tilde{\mathbb{P}}(A^c) \leq 4 \frac{\tilde{\text{Var}}[X]}{\tilde{\mathbb{E}}[X]^2}$
- ▶ Estimate $\text{Var}[X]$, $\tilde{\mathbb{E}}[X]$ (2nd moment) and $\tilde{\text{Var}}[X]$ (3rd moment)

We take X as a **manageable proxy** of $Z - 1$ restricting its **chaos expansion** to

$$I = \{(n_1, x_1), \dots, (n_k, x_k)\} \quad \text{with} \quad \begin{cases} \text{width}(I) = n_k - n_1 \leq \varepsilon N, \\ |I| = k \leq \log(\varepsilon N) \end{cases}$$

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Conclusions

Solution $u_N(t, x)$ of discretized 2D SHE $\longleftrightarrow$ directed polymer partition function

► Critical regime: $u_N \xrightarrow{N \rightarrow \infty} \text{SHF}^\vartheta$ independent of $\xi = \lim_{N \rightarrow \infty} \xi_N$

$\rightsquigarrow$ Enhanced Noise Sensitivity via influences

► Supercritical regime (strong disorder): local extinction $u_N \rightarrow 0$

$\rightsquigarrow$ Mass escape on the spatial scale $\exp(c e^\vartheta) = \exp(c N e^{-\pi/\beta^2})$

Grazie